Experiment 12:-

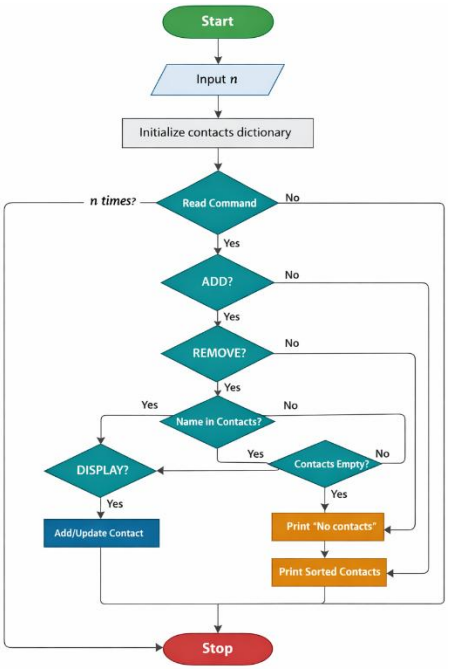
Problem Statement:-

Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.

Algorithm:-

1. Start
2. Input number of operations `n`
3. Create an empty dictionary `contacts`
4. Repeat `n` times:
    - Read command input
    - If command is `"ADD"`:
        - Take `name` and `phone`
        - Add/update in `contacts`
    - Else if command is `"REMOVE"`:
        - Take `name`
        - If `name` exists in `contacts`, delete it
    - Else if command is `"DISPLAY"`:
        - If `contacts` is empty, print `"No contacts"`
        - Else print all contacts in sorted order of names
5. Stop

Flowchart:-



Execution:-

CODETANTRA
Home
arjun.gahane.batch2025@sitnagpur.siu.edu.in
Support
Logout
12.1.1. Phone Number Database Management
02:55
Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.
The program should support three operations: adding a contact (if the name already exists, update the phone number), removing a contact (if the name does not exist, ignore the operation), and displaying all contacts in alphabetical order by name.
Input Format:
1. First line contains an integer n representing the number of operations.
2. Each of the next n lines contains a single operation in one of the following formats:
• "ADD <name> <phone_number>" where <name> is a string, <phone_number> is a string of 10 digits.
• "REMOVE <name>" - name is a string.
• "DISPLAY" - no additional parameters.
Output Format:
• For display operation, print all contacts in alphabetical order by name in the format:
Explorer
phoneDat...
Submit
Debugger
# Phone Number Database Management
n = int(input())
contacts = {}
for _ in range(n):
command = input().split()
if command[0] == "ADD":
name = command[1]
phone = command[2]
contacts[name] = phone # add or update
2
ADD Alice 7771112223
ADD bob 98465879987
=== YOUR PROGRAM HAS ENDED ===